Crown and bridge (fixed prosthesis)

Lecture two

Dr. Mondale R. Hadi

Indications, types and steps of crown construction

Indications for crown Construction:

Purposes of making crown and bridge restorations:

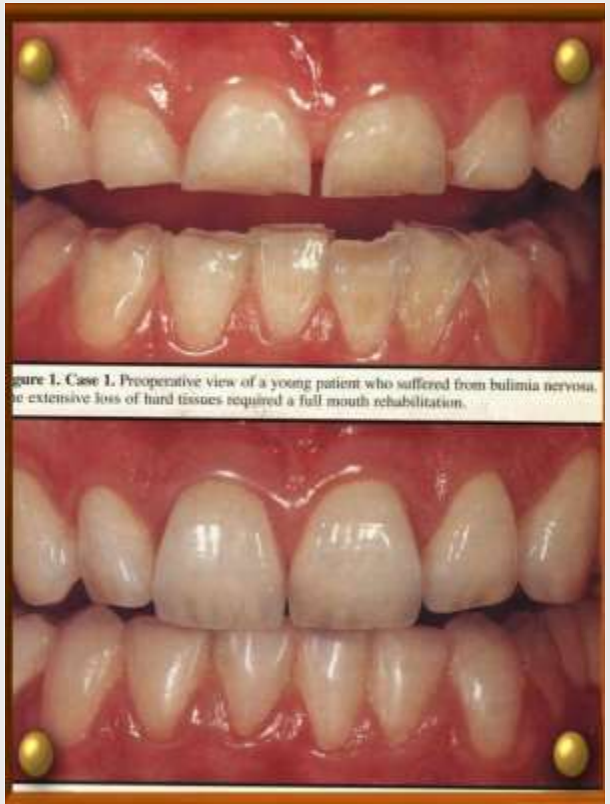
1-To restore the grossly damaged, fractured or teeth with a heavy filling [amalgam or composite].

2-To restore masticatory function and speech.

3- To restore the esthetic (hypoplastic condition whether heredity defect or acquired defect).

4- To maintain the periodontal health by recontouring The occlusion And prevents food impaction.

5- As a retainer for the bridge .



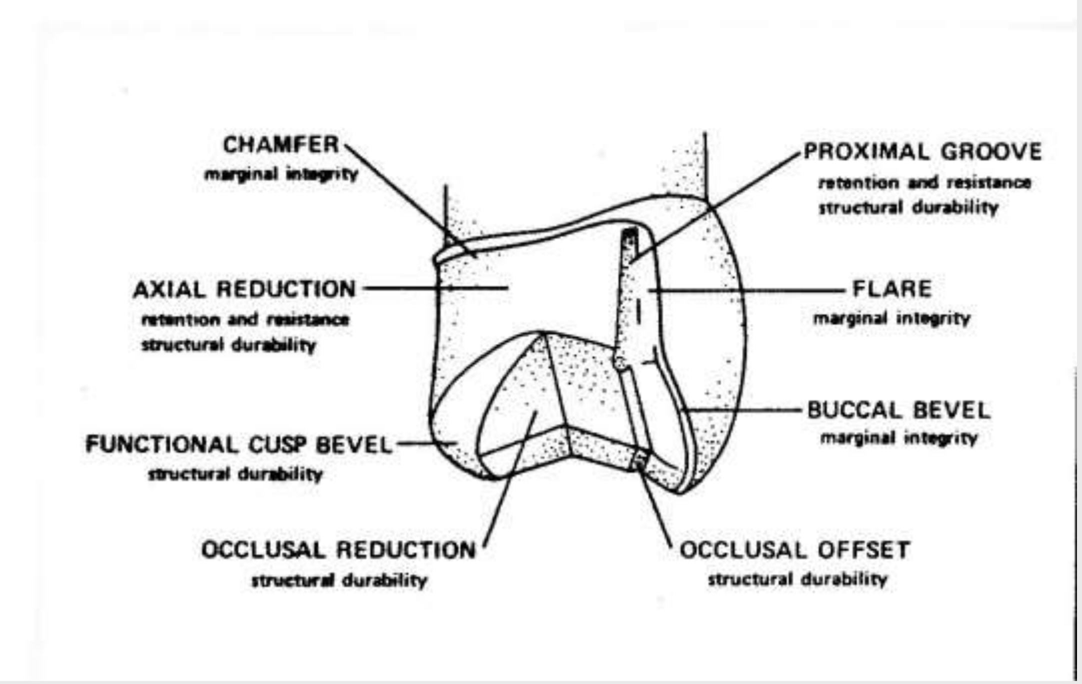
Classification of crowns

I. According to coverage area

Complete crown : It is the crown that covers all the coronal portion of the tooth ,such as full metal crown ,Jacket crown which is a complete crown made of non-metallic materials plastic or ceramic.

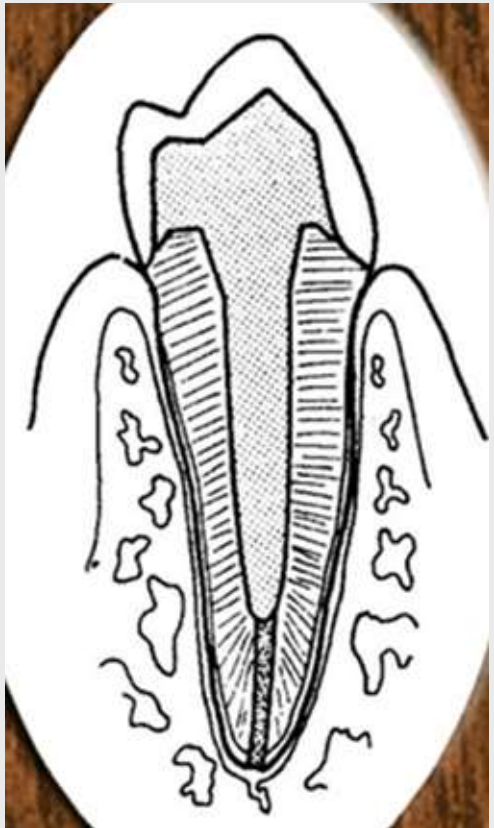
Partial Crown :

It is a crown that covers part of the coronal portion of the tooth such as 3/4 crown , 7/8Crown.



Post crown:

Crown with complete replacement of the coronal portion Of the tooth. This type of crown retains itself by means of a post extended inside the root canal of the tooth.



II. According to the materials used in the fabrication of the crown:

1. Metal crown: made from gold alloy and its alternatives such as full metal crown and 3/4 crown.

2. Non- metal crown: made from acrylic resin, zirconium or porcelain as in jacket crown.

3. A combination of metal and plastic materials such as porcelain fused to metal crown.








METAL CERAMIC
CROWNS

Steps in the construction of cast restorations

1. Diagnosis.
2. Tooth preparation.
3. Final impression.
4. Temporary restoration (Provisional restoration).
5. Construction of working model.
6. Waxing.

Steps in the construction of cast restorations

7. Investing.

8. Casting.

9. Cleaning and finishing.

10. Ceramic application

11. Try-in and cementation.

Steps in the construction of cast restorations:

Note: Steps (1-4, and 11) are clinical steps, while steps (5-10) are laboratory steps carried out in the lab by the laboratory technician.

Note: The steps mentioned above concern the fabrication of cast restorations which are restorations made entirely from metal or a combination of metal and plastic material. All ceramic restorations are fabricated using other laboratory procedures such as CAD/CAM (Computer Aided Design / Computer Aided Manufacturing).

Steps in crown construction

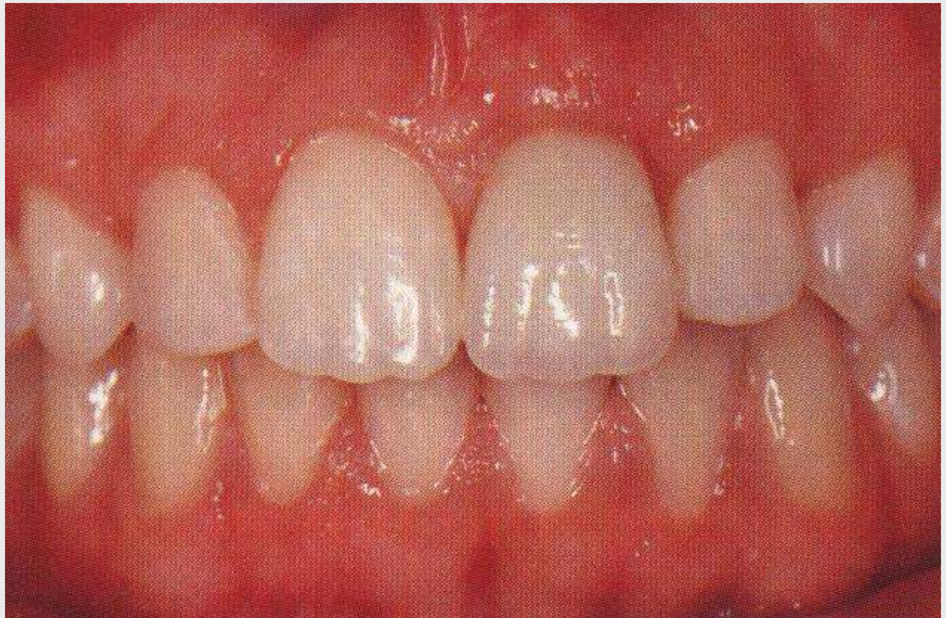
1- Diagnosis :

The first step to be sure that the case is indicated for fixed prosthesis construction. This is decided after a thorough examination of the teeth and surrounding structures, which includes:

Clinical examination:

A) Periodontal examination :

Proper oral hygiene Should be available to ensure That no plaque accumulation is formed on the crown margins which might lead to caries .



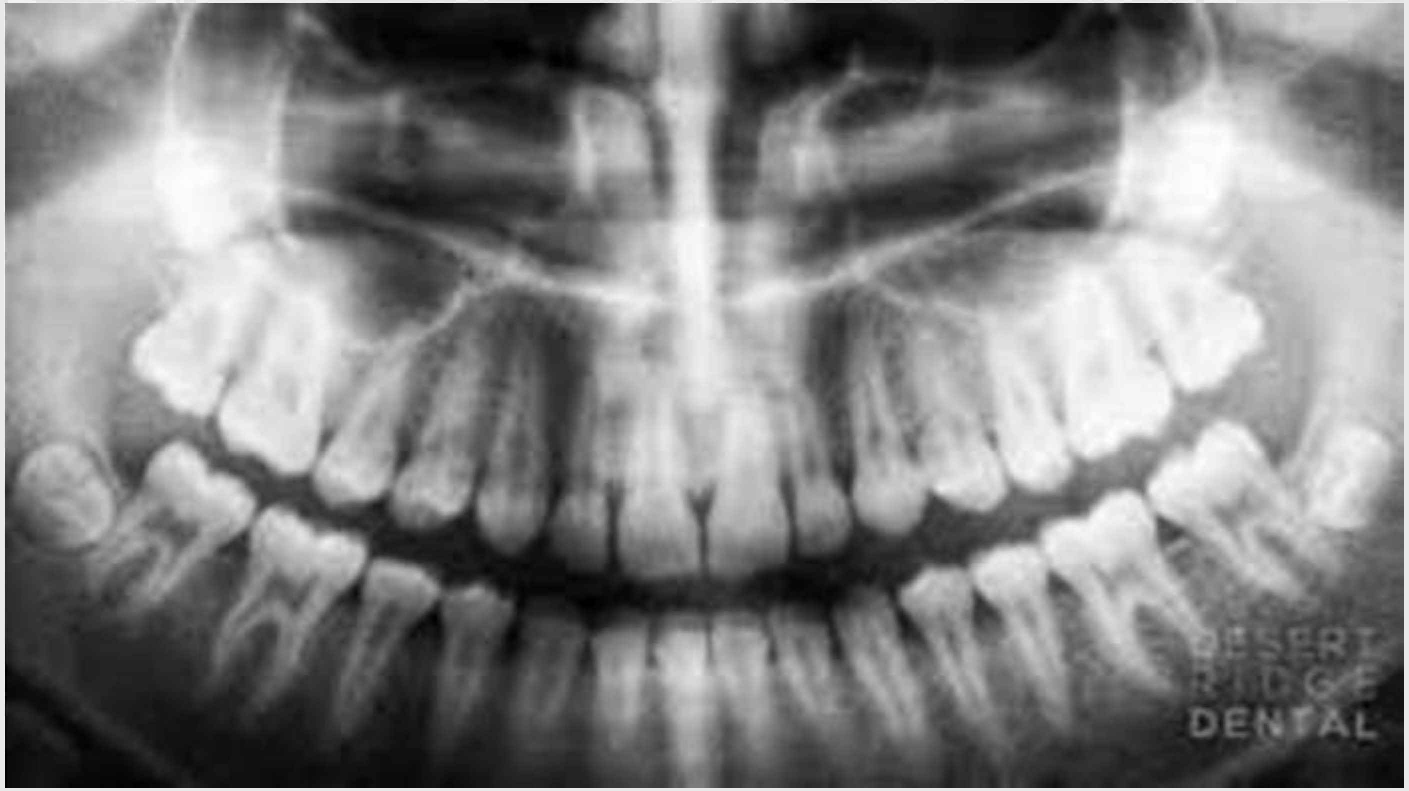
B: Dental Examination:

(1) Visual examination :

We should examine the occlusion, crowding ,spacing, rotation, the condition of remaining tooth structure and future treatment is also analyzed .

(2)Radiographic: The radiograph reveals the shape and number of the roots, the condition of the surrounding structures, and the bone support of the tooth (crown/root ratio). The ideal crown/root ratio of a tooth to be used as an abutment for fixed partial denture is 1:2.


Dark Area Below
the Tooth Enamel.


DESERT
RIDGE
DENTAL

The radiograph also reveals the presence of a lesion in the bone, root canal treatment, fracture in the tooth or root, bone loss, unerupted teeth…… etc . These informations will affect on the prognosis of the treatment.

2-Tooth Preparation :

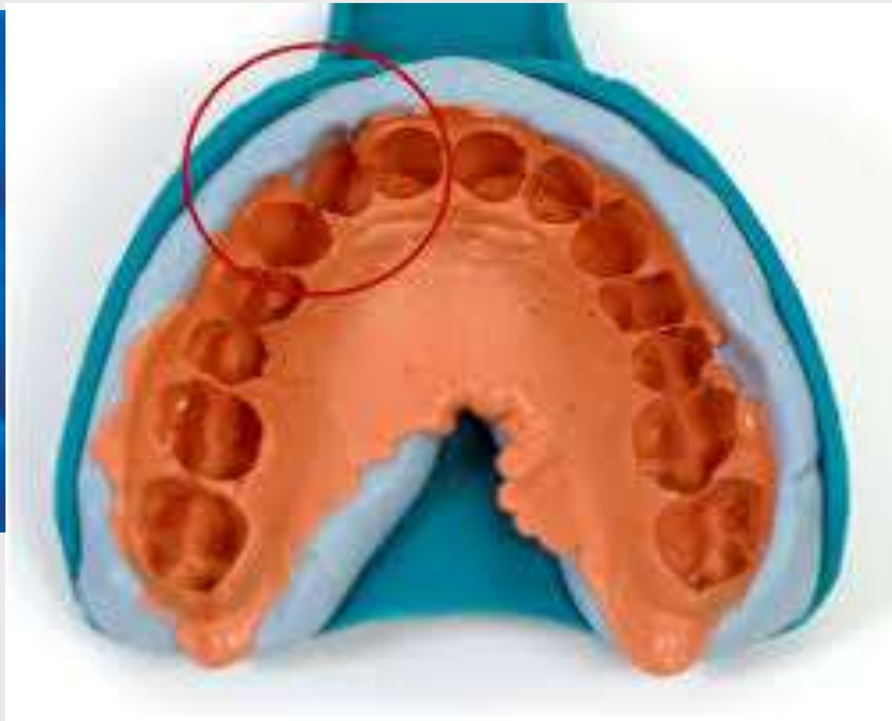
It is the cutting or instrumentation of the abutment tooth to be in a form that makes seating of a crown on it possible.

Prepared tooth(abutment): **It's the final form of a tooth after cutting or preparation procedure.**

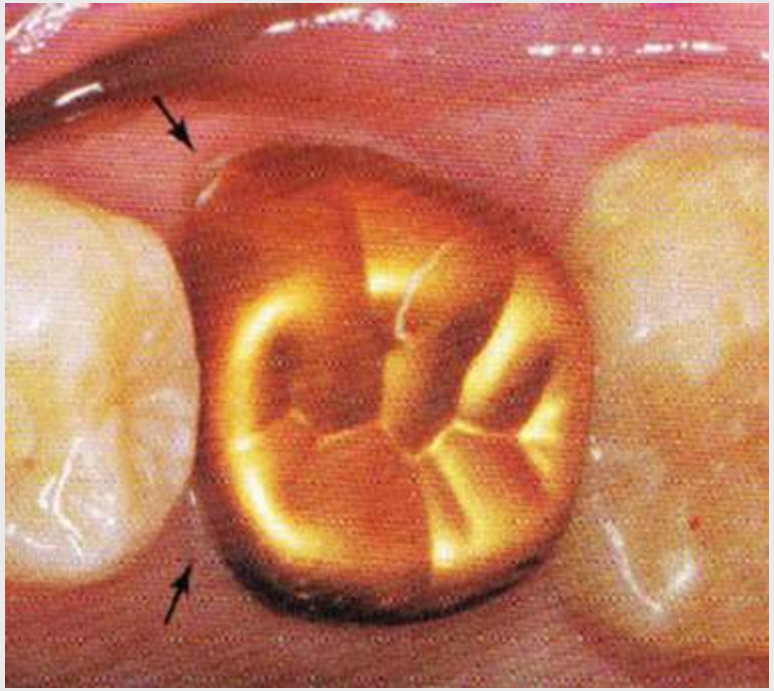
rotary instruments are used to reduce the height and contour of the tooth.

The tooth is prepared so that the crown restoration can slide into place and be able to withstand the forces of occlusion.

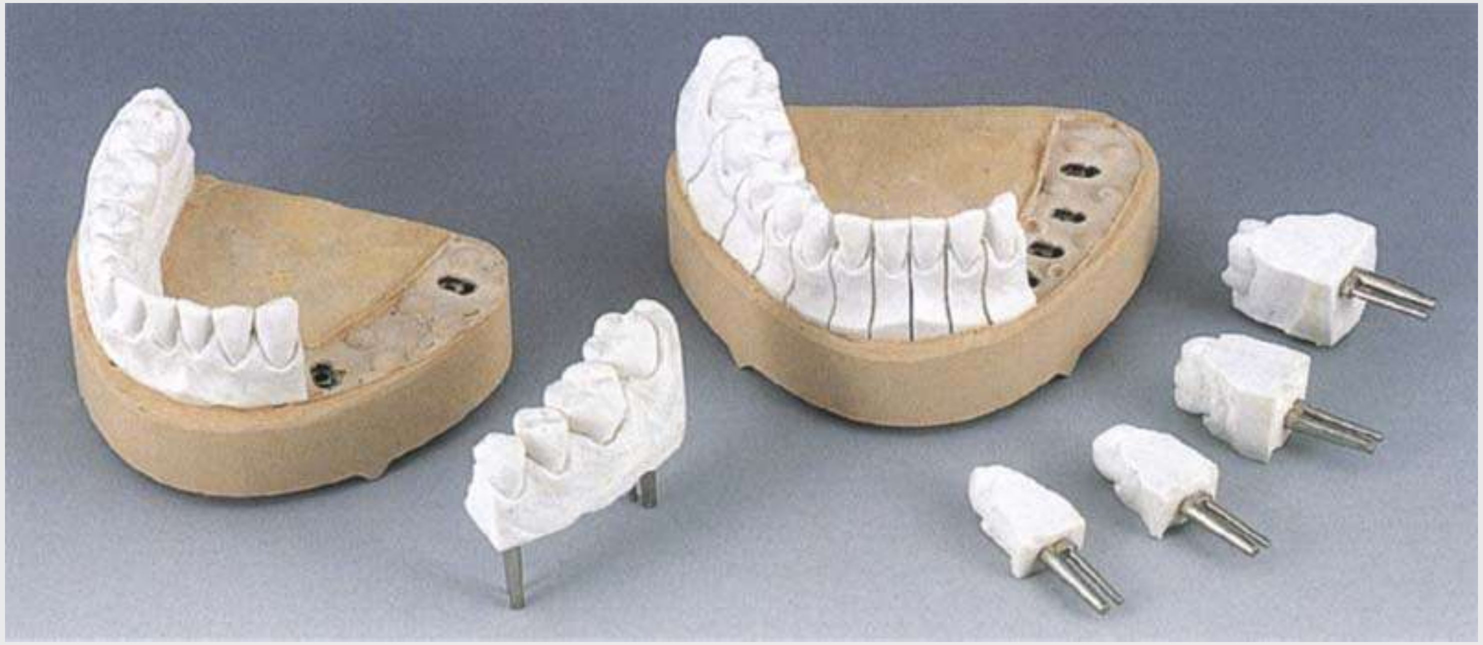
3.Final impression and bite registration.



4- Temporary restorations



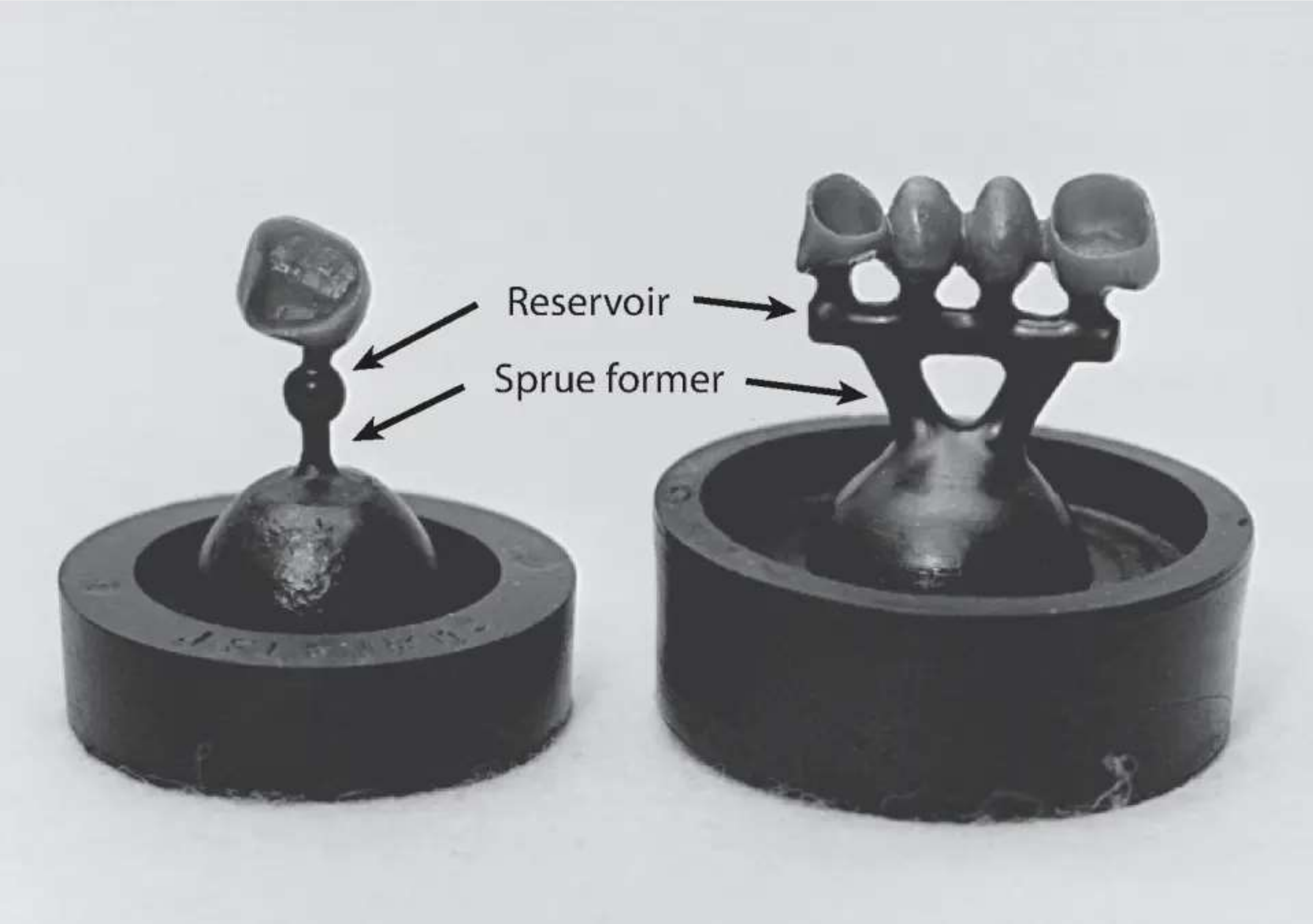
5-Working models and cast mounting



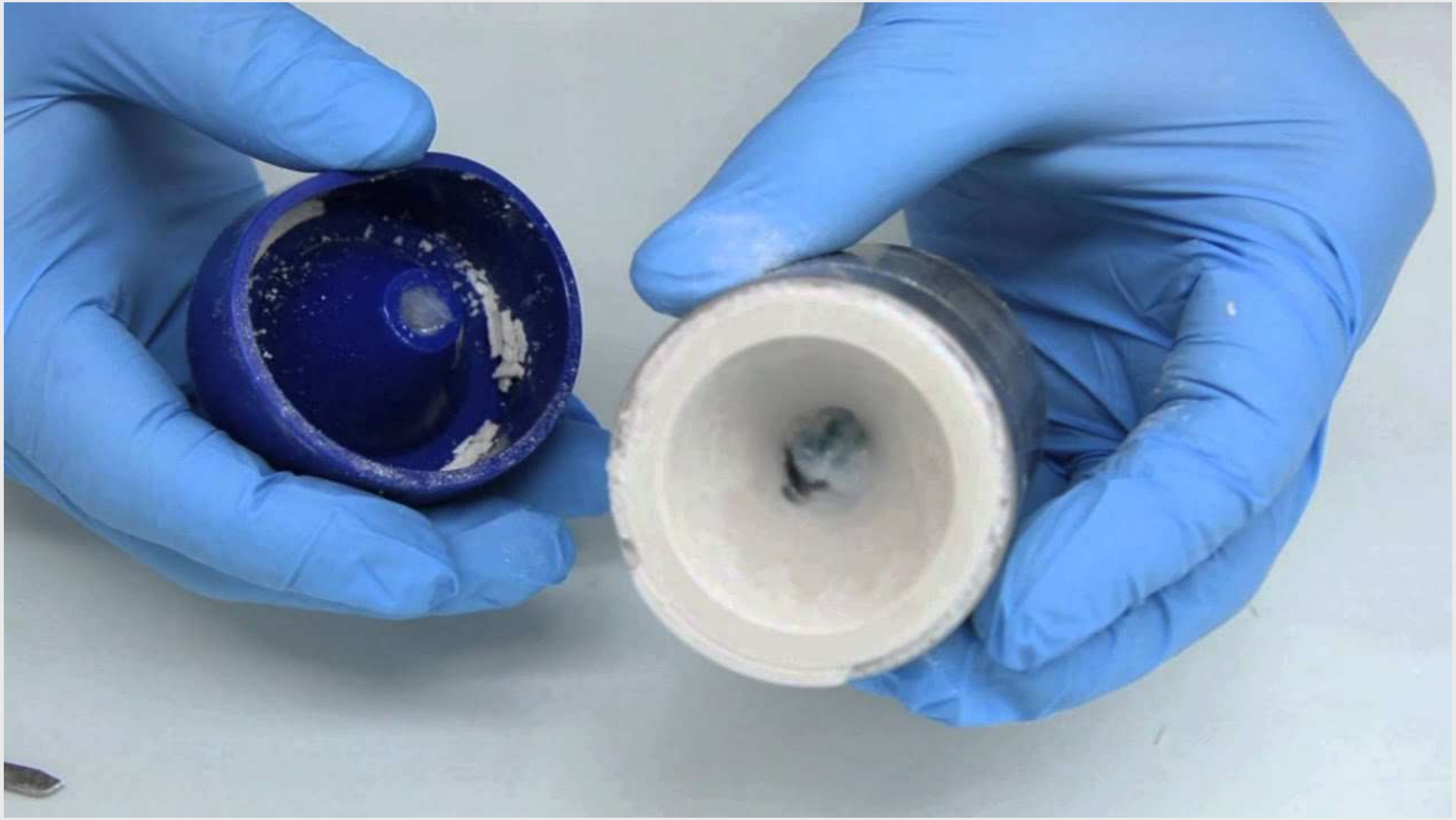
6-waxing

7-Investing (wax elimination)

7-Investing (wax elimination)

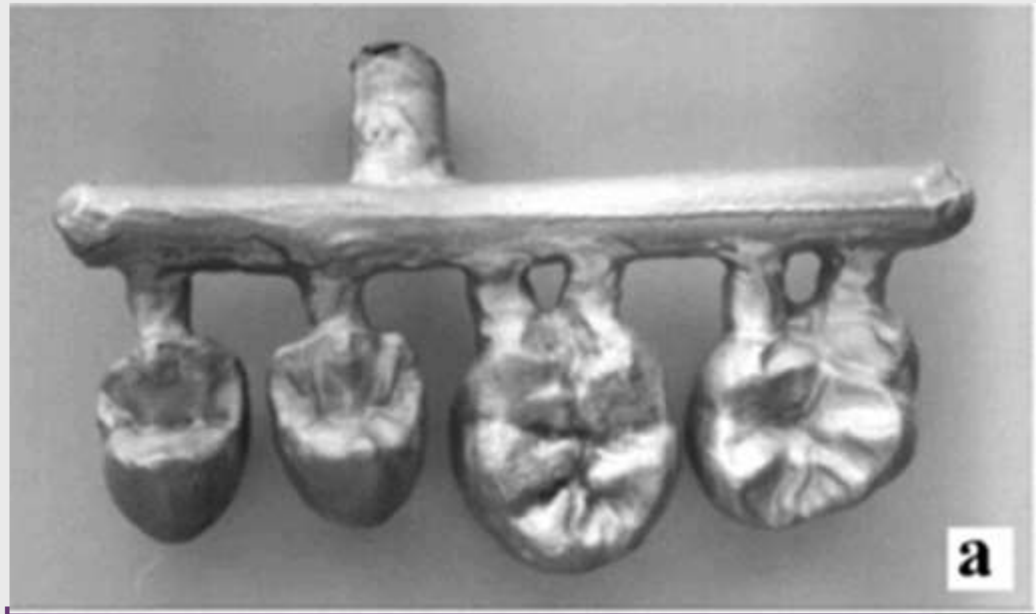


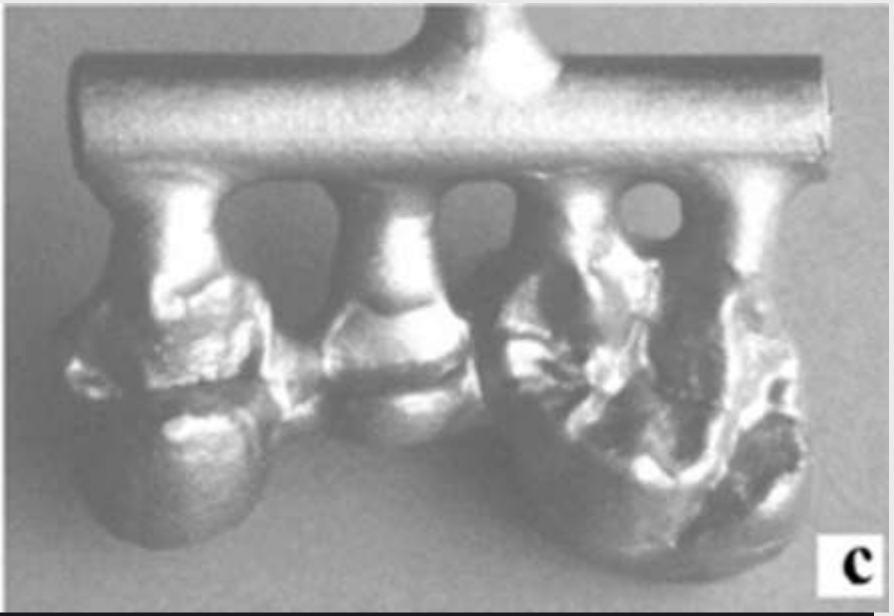
7-Investing (wax elimination)



8- casting

8- casting

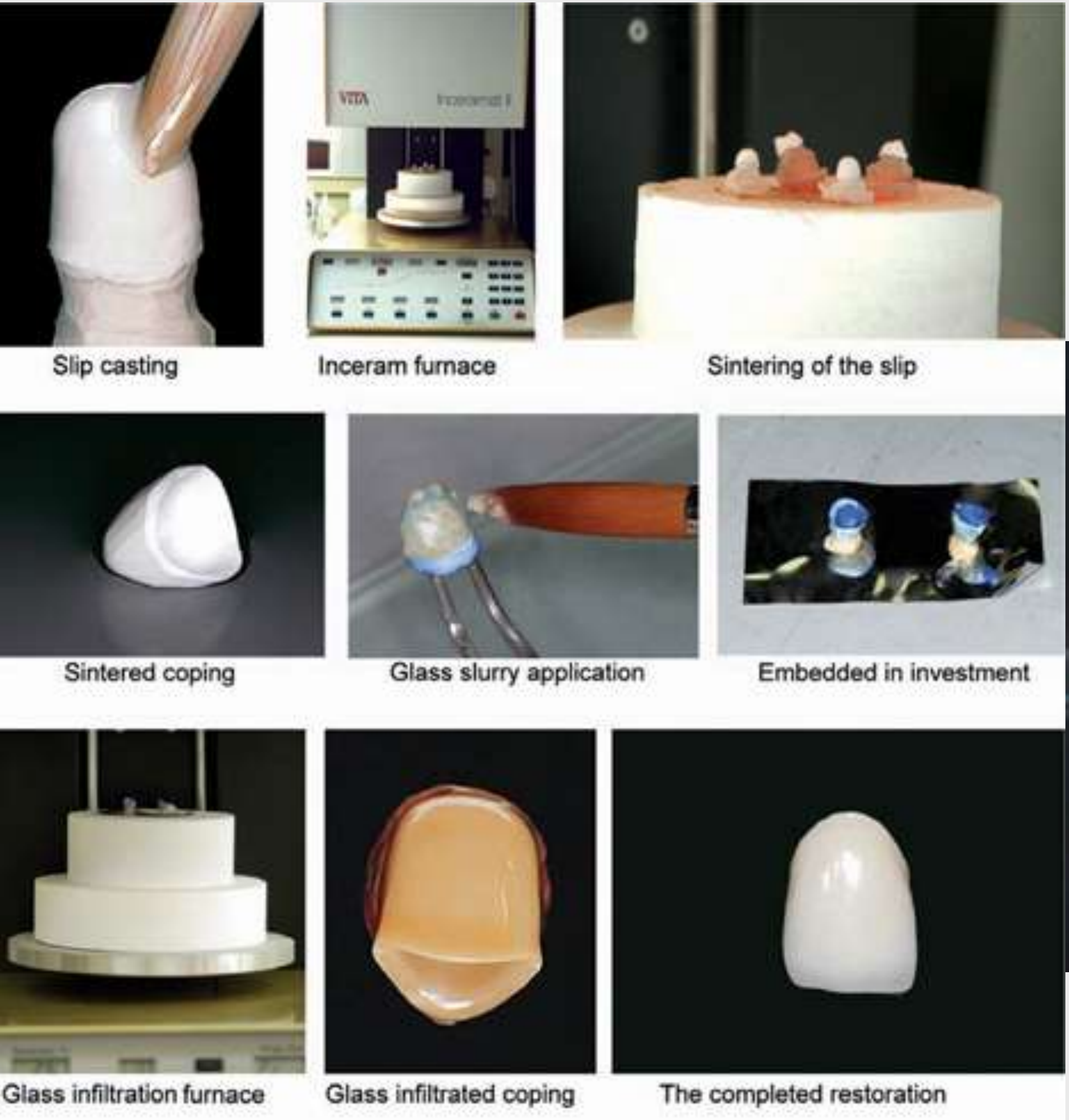




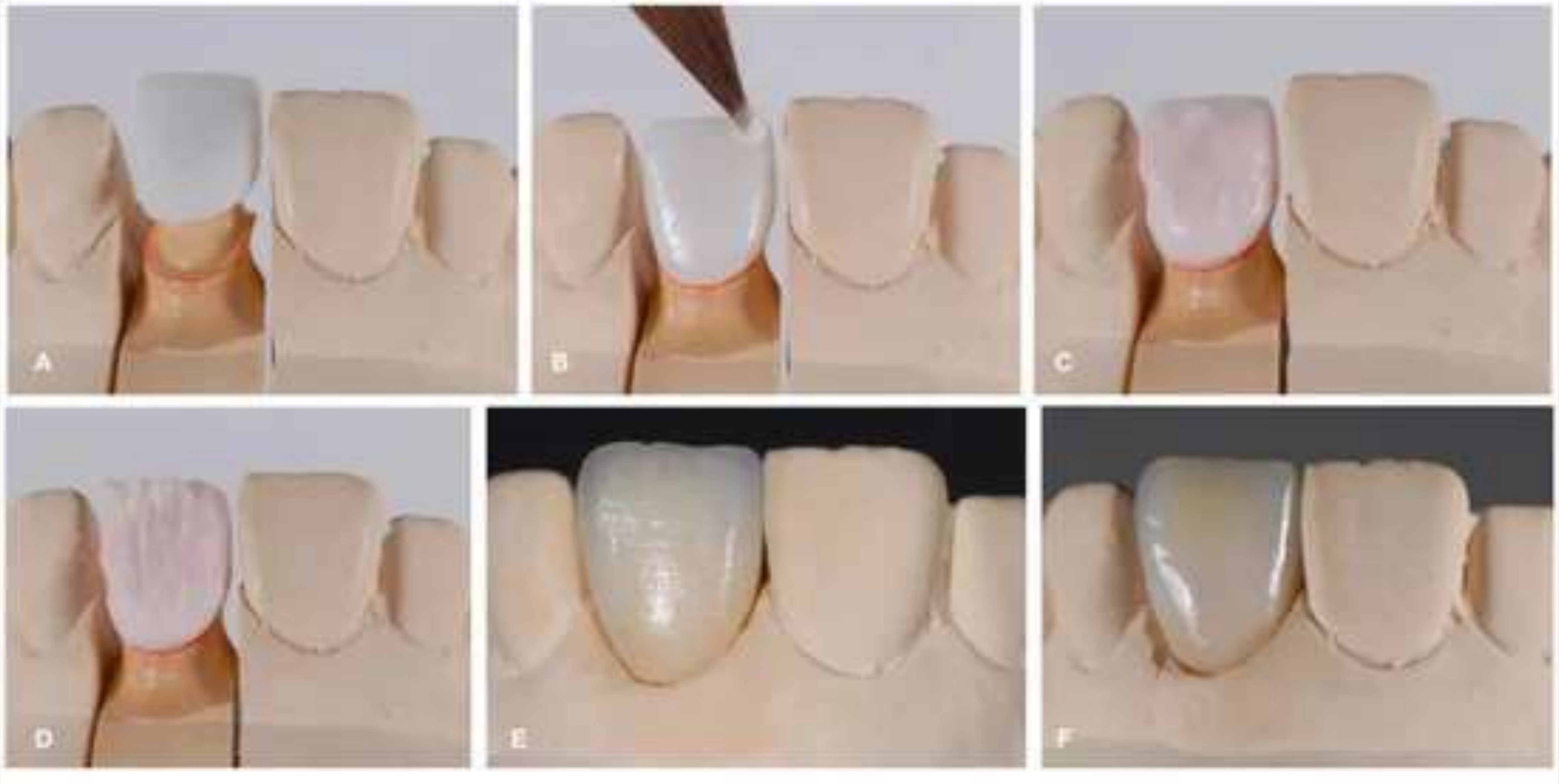
9- Finishing and polishing

10- ceramic facing for metal-ceramic restorations

Fabrication of full-ceramic restorations



Ceramic fabrication for full ceramic restoration:



11- cementation

Disadvantages Of crowns:-

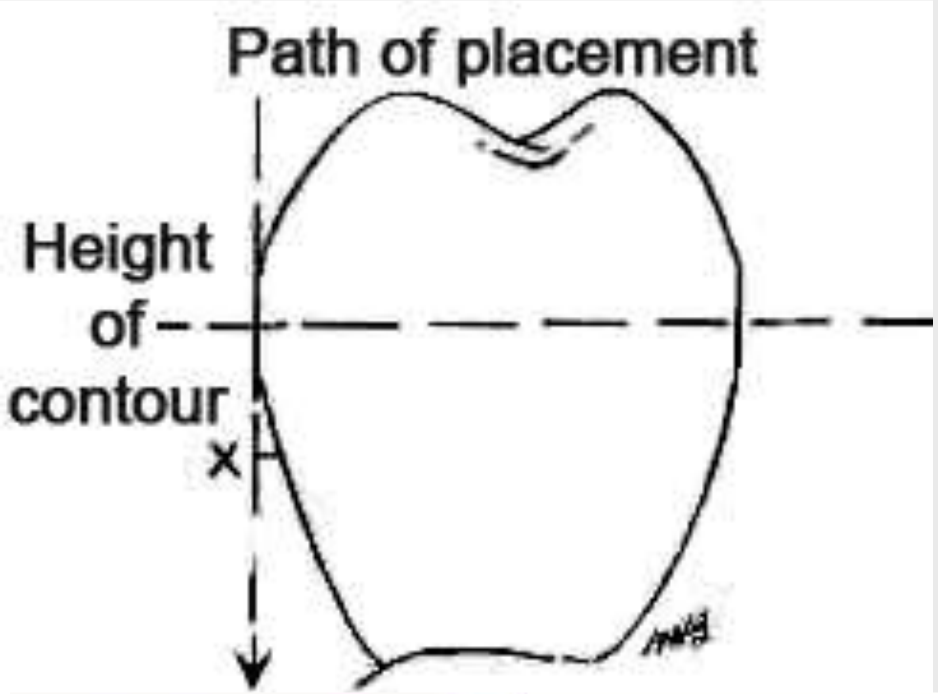
1-Heat generation due to preparation of the teeth , this heat generation might affect the health of the pulp.

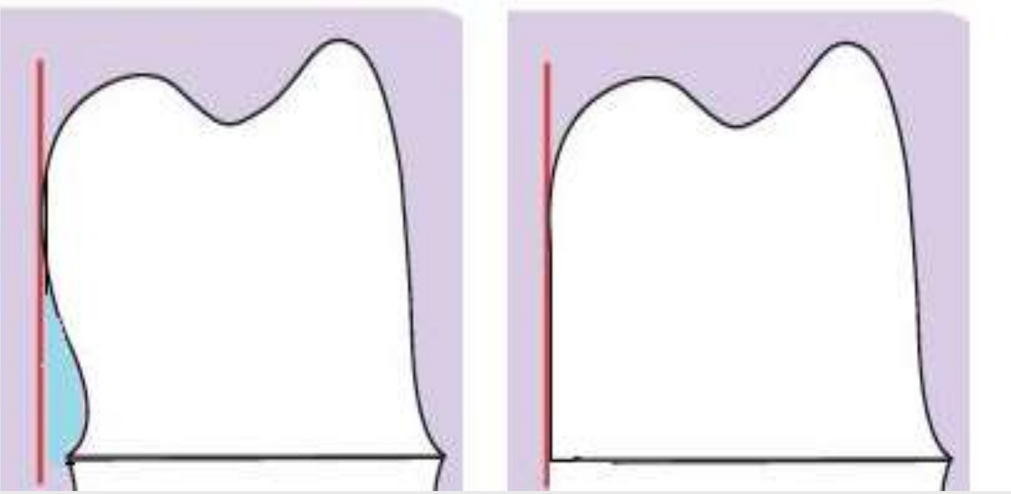
2-Periodontal problems ,food Impaction , and secondary caries might develop.

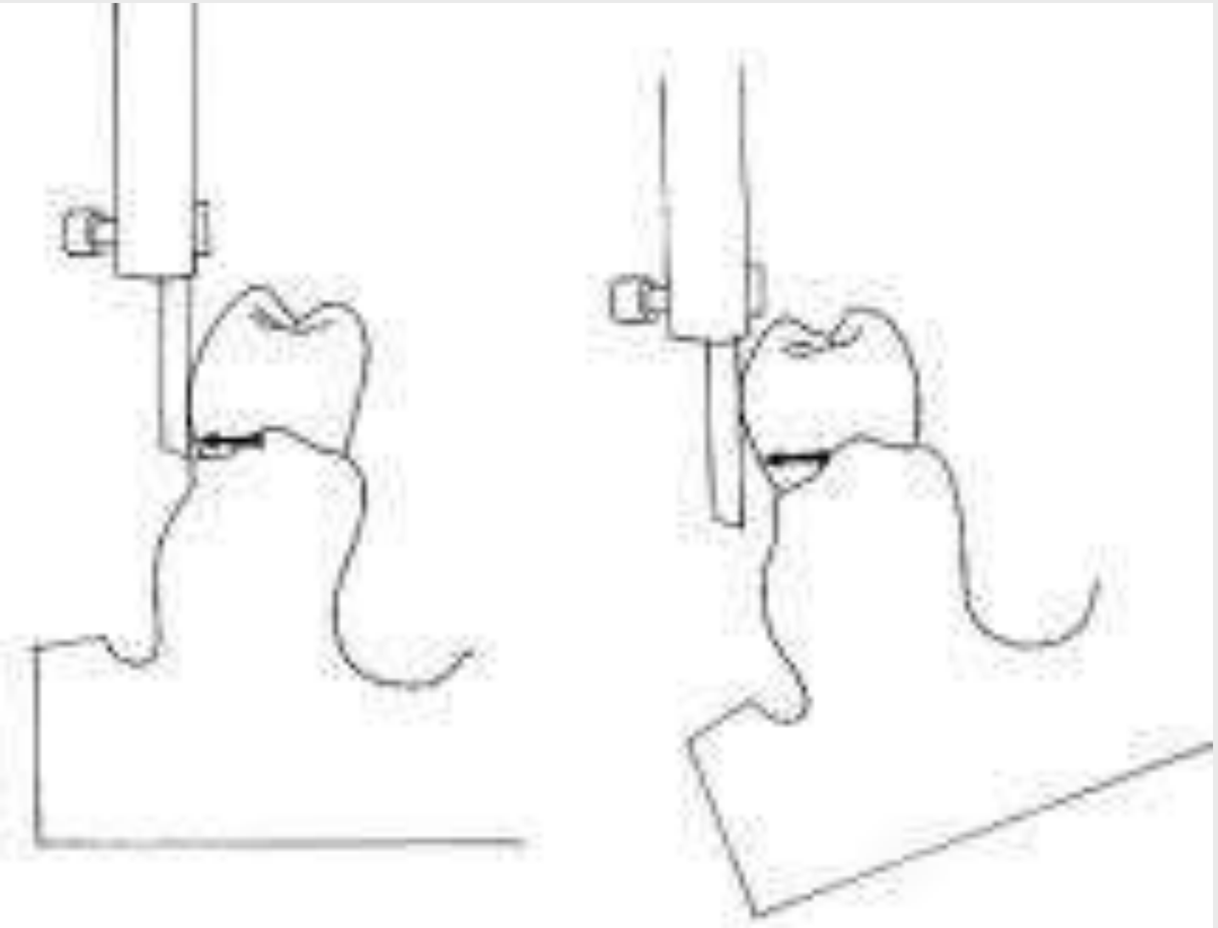
3-Over preparation can cause pulp irritation which might lead to death of the pulp, Therefore water coolant must be used during preparation procedure. Excessive tooth preparation can also weaken tooth structure .

Objectives of tooth Preparation:-

1-To eliminate undercuts from the axial surfaces of the tooth .





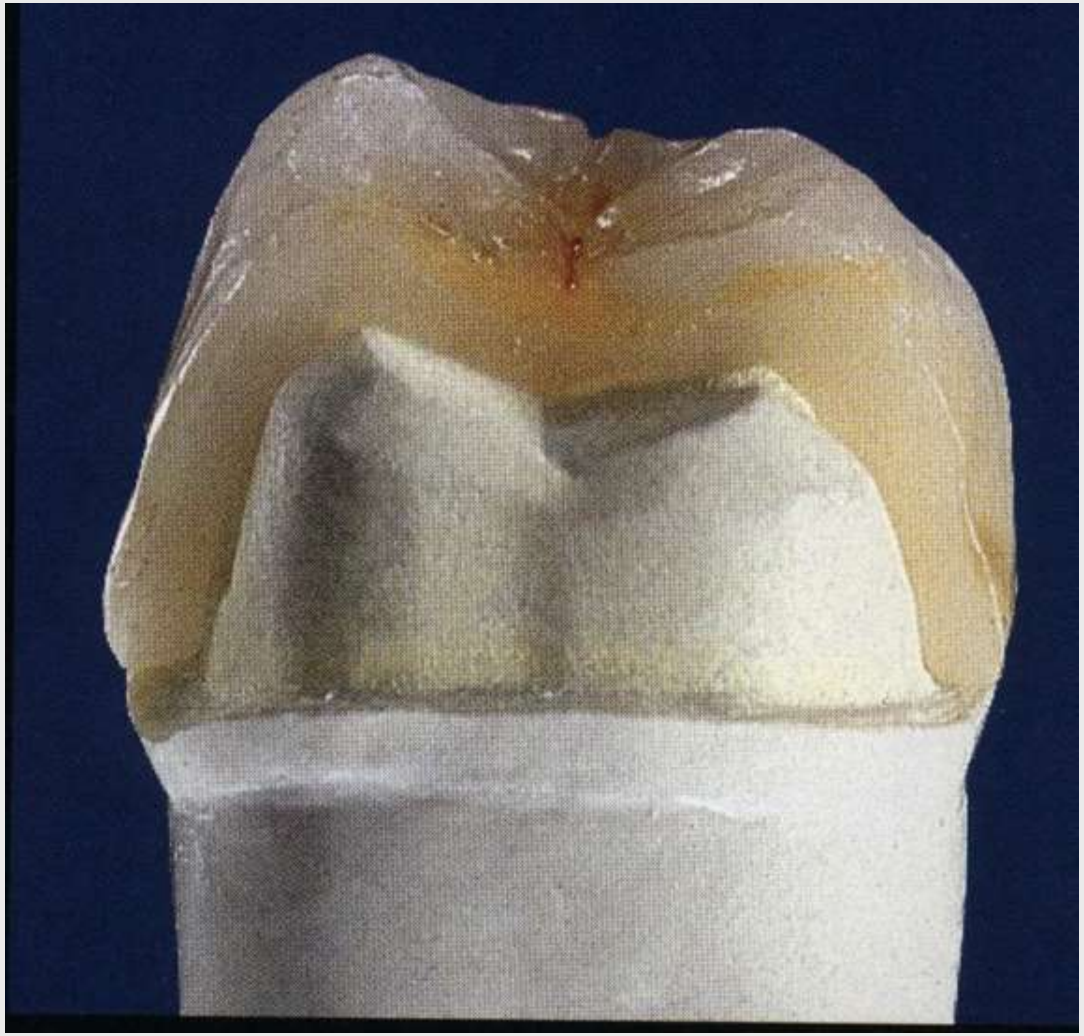


2-To provide enough space for the crown restoration to withstand the force of mastication without disturbing the normal occlusion of the patient.

This space depends on the material used, so the metal material needs little space while the plastic (or ceramic) material needs more space .

3-Not to enlarge the size of the tooth.

4-To provide good esthetic.



Any question

Thank you